Statistics for Data Science - 2

Week 2 Graded Assignment

Multiple random variables

1. Let X and Y be two independent discrete random variables with CDFs F_X and F_Y, respectively. Define another random variable $Z = \min(X, Y)$, then the CDF of Z is

   a) $\min(F_X, F_Y)$

   b) $F_X F_Y$

   c) $F_X + F_Y + F_X F_Y$

   d) $F_X + F_Y - F_X F_Y$

   Solution:

$$\begin{aligned} F_Z(z) = P(Z \leq z) &= P(\min(X, Y) \leq z) \\ &= 1 - P(\min(X, Y) > z) \\ &= 1 - P(X > z, Y > z) \end{aligned}$$

   Since X and Y are two independent discrete random variables,
   $P(X > z, Y > z) = P(X > z)P(Y > z)$

$$\begin{aligned} F_Z(z) &= 1 - P(X > z)P(Y > z) \\ &= 1 - [(1 - P(X \leq z))(1 - P(Y \leq z))] \\ &= 1 - [(1 - F_X(z))(1 - F_Y(z))] \\ &= F_X(z) + F_Y(z) - F_X(z)F_Y(z) \end{aligned}$$

2. Let X and Y be two independent random variables with PMFs

$$f_X(k) = f_Y(k) = \begin{cases} \frac{1}{6} & \text{for } k = 1, 2, 3, 4, 5, 6. \\ 0 & \text{otherwise} \end{cases}$$

   Define $Z = X - Y$. Find the value of $f_Z(3)$.

   a) $\dfrac{4}{12}$

   b) $\dfrac{3}{12}$

   c) $\dfrac{5}{12}$

d) $\frac{1}{12}$

Solution:

$$\begin{aligned} f_Z(3) = P(Z=3) = P(X-Y=3) \\ &= P(X=4, Y=1) + P(X=5, Y=2) + P(X=6, Y=3) \end{aligned}$$

Given that X and Y are two independent random variables.
$\Rightarrow P(X=x, Y=y) = P(X=x)P(Y=y)$ for all (x, y).

$$\begin{aligned} f_Z(3) &= P(X=4, Y=1) + P(X=5, Y=2) + P(X=6, Y=3) \\ &= P(X=4)P(Y=1) + P(X=5)P(Y=2) + P(X=6)P(Y=3) \\ &= \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \\ &= \frac{3}{36} \\ &= \frac{1}{12} \end{aligned}$$

3. Let $X \sim$ Geometric(p) and $Y \sim$ Geometric(p) be independent and let $Z = X + Y$. Determine the values of p for which $P(Z = 26) > P(Z = 25)$.

   a) $p > 0.02$
   b) $p < 0.04$
   c) $p > 0.15$
   d) $p < 0.30$
   e) $p = 0.05$

Solution:
If $X \sim$ Geometric(p) and $Y \sim$ Geometric(p) are two independent random variables and $Z = X + Y$, then

$$P(Z = n) = (n-1)p^2(1-p)^{n-2}$$

(Try derivation yourself)

We have to find the value of p for which $P(Z = 26) > P(Z = 25)$.

$P(Z = 26) = (26-1)p^2(1-p)^{26-2}$
and $P(Z = 25) = (25-1)p^2(1-p)^{25-2}$

Comparing both, we will get
$25p^2(1-p)^{24} > 24p^2(1-p)^{23}$

$\Rightarrow 25(1-p) > 24$
$\Rightarrow 1-p > \frac{24}{25}$
$\Rightarrow p < 0.04$

4. The following options gives the joint PMF of the random variables X and Y. If the random variables X and Y are independent, then which of the following option(s) can be the joint PMF of X and Y?

$X \backslash Y$	0	1	2
0	0.01	0	0
1	0.09	0.09	0
2	0	0	0.81

a)

$X \backslash Y$	0	1	2
0	0.06	0.18	0.12
1	0.04	0.12	0.48

b)

$X \backslash Y$	0	1	2
0	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$
1	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$
2	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$

c)

Y / X	0	1	2
0	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$
1	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{3}{10}$

d)

Y / X	0	1
0	0.10	0.15
1	0.20	0.30
2	0.10	0.15

e)

Answer: e

Solution:

In option a)
$P(X=0, Y=1) = 0$ but $P(X=0) = 0.01+0+0 = 0.01$ and $P(Y=1) = 0+0.09+0 = 0.09$
$\Rightarrow P(X=0, Y=1) \neq P(X=0)P(Y=1)$
Therefore, option (a) cannot be the joint PMF of X and Y.

In option b)
$P(X=0, Y=0) = 0.06$ but $P(X=0) = 0.06 + 0.18 + 0.12 = 0.36$ and $P(Y=0) = 0.06 + 0.04 = 0.10$
$\Rightarrow P(X=0, Y=0) = 0.06 \neq 0.036 = P(X=0)P(Y=0)$
Therefore, option (b) cannot be the joint PMF of X and Y.

In option c)
$P(X=1, Y=0) = 1/6$ but $P(X=1) = 1/6 + 1/12 + 1/8 = 3/8$ and $P(Y=0) = 1/12 + 1/6 + 1/4 = 1/2$
$\Rightarrow P(X=1, Y=0) = 1/6 \neq 3/16 = P(X=1)P(Y=0)$
Therefore, option (c) cannot be the joint PMF of X and Y.

In option d)
$P(X=0,Y=1)=1/5$ but $P(X=0)=1/10+1/5+1/5=1/2$ and $P(Y=1)=1/5+1/10=3/10$
$\Rightarrow P(X=0,Y=1)=1/5\neq 3/20=P(X=0)P(Y=1)$
Therefore, option (d) cannot be the joint PMF of X and Y.

In option e)
For every $(x,y), P(X=x,Y=y)=P(X=x)P(Y=y)$ (check yourself)
Hence option (e) is the joint PMF of X and Y.

5. Let X and Y be two independent random variables such that $X\sim$ Bernoulli(0.2) and $Y\sim$ Bernoulli(0.4). Let another random variable Z be defined as $Z=X+Y$. Find the value of $f_{X|Z=1}(1)$. Enter the answer correct to two decimal places.
Answer: [0.25, 0.29]

Solution:

$X\sim$ Bernoulli(0.2)
$Y\sim$ Bernoulli(0.4)

$Z=X+Y$

$$\begin{aligned}
f_{X|Z=1}(1) =& \frac{P(X=1,Z=1)}{P(Z=1)}\\
=& \frac{P(X=1,Y=0)}{P(X=1,Y=0)+P(X=0,Y=1)}\\
=& \frac{P(X=1)P(Y=0)}{P(X=1)P(Y=0)+P(X=0)P(Y=1)}\\
=& \frac{0.2\times 0.6}{(0.2\times 0.6)+(0.8\times 0.4)}\\
=& 0.28
\end{aligned}$$

6. Let X_1, X_2 and X_3 be three independent and identically distributed Poisson random variables with $\lambda_i=3$ for all i. Find the probability that exactly one of the X_i equals 0 and exactly one of the X_i equals 1?

(a) $18e^{-6}(1-4e^{-3})$
(b) $9e^{-6}(1-4e^{-3})$
(c) $3e^{-6}(1-4e^{-3})$
(d) $12e^{-6}(1-4e^{-3})$

Solution:

First we will find the probability such that $X_1 = 0, X_2 = 1$ and X_3 do not take values 0 and 1.
Since all the four random variable are independent, we have
$P(X_1 = 0, X_2 = 1, X_3 \neq \{0, 1\}) = P(X_1 = 0)P(X_2 = 1)P(X_3 \neq \{0, 1\})$

Now, $P(X_1 = 0) = e^{-3}, P(X_1 = 1) = 3e^{-3}$

$$\begin{aligned} P(X_3 \neq \{0,1\}) &= 1 - P(X_3 = \{0,1\}) \\ &= 1 - [P(X_3 = 0) + P(X_3 = 1)] \\ &= 1 - 4e^{-3} \end{aligned}$$

Now, $P(X_1 = 0, X_2 = 1, X_3 \neq \{0, 1\}) = e^{-3}3e^{-3}(1 - 4e^{-3}) = 3e^{-6}(1 - 4e^{-3})$
We can choose such pairs of X_i in 3! ways.
Therefore, probability that exactlyX_i equals 0 and exactly one X_i equals 1 is given by $6 \times 3e^{-6}(1 - 4e^{-3}) = 18e^{-6}(1 - 4e^{-3})$ ways.

7. Let $X \sim$ Bernoulli(0.3) and $Y \sim$ Bernoulli(0.5) be independent. Define $Z = X+Y-XY$, find the distribution of Z.

   (a) $Z \sim$ Bernoulli(0.35)
   (b) $Z \sim$ Bernoulli(0.65)
   (c) $Z \sim$ Bernoulli(0.15)
   (d) $Z \sim$ Bernoulli(0.85)

Solution:

Given $X \sim$ Bernoulli(0.3) and $Y \sim$ Bernoulli(0.5) are independent.
$Z = X + Y - XY$

Since $X \sim$ Bernoulli(0.3), it will take the values in $\{0, 1\}$.
Similarly Y will take values in $\{0, 1\}$.

Also X and Y are given to be independent, therefore $f_{XY}(x, y) = f_X(x)f_Y(y)$ for all (x, y).

Consider the following joint distribution table of X and Y.

X	Y	$X+Y-XY$	$f_{XY}(x,y)$
1	1	1	$(0.3)(0.5) = 0.15$
1	0	1	$(0.3)(0.5) = 0.15$
0	1	1	$(0.7)(0.5) = 0.35$
0	0	0	$(0.7)(0.5) = 0.35$

The range of Z is $\{0, 1\}$.
$P(Z = 0) = 0.35, P(Z = 1) = 0.65$

Therefore, $Z \sim$ Bernoulli(0.65).

8. Let X and Y be independent and identically distributed Geometric random variables with parameter 0.6.

   (a) Find $P(X = 2 \mid X + Y = 11)$. Enter the answer correct to one decimal place.
   Answer: 0.1

   Solution:
   Given $X, Y \sim$ i.i.d. Geometric(0.6).

$$\begin{aligned} P(X = 2 \mid X + Y = 11) &= \frac{P(X = 2, X + Y = 11)}{P(X + Y = 11)} \\ &= \frac{P(X = 2)P(Y = 9)}{\sum\limits_{i=1}^{10} P(X = i, Y = 11 - i)} \\ &= \frac{(0.4)(0.6) \times (0.4)^8(0.6)}{\sum\limits_{i=1}^{10} P(X = i)P(Y = 11 - i)} \\ &= \frac{(0.4)^9(0.6)^2}{10(0.4)^9(0.6)^2} = 0.1 \end{aligned}$$

   (b) Find $P(X = 7 \mid X + Y = 11)$. Enter the answer correct to one decimal place.
   Answer: 0.1

   Solution:
   This can also be solved using the above steps.
   Remark: For any i, $P(X = i \mid X + Y = n) = \dfrac{1}{n-1}$, for $i : 1 \to n - 1$.

9. The joint distribution of X and Y is given by

$$f_{XY}(x, y) = \frac{9}{16 \times 4^{x+y}},$$

   where $T_X, T_Y \in \{0, 1, 2, \ldots\}$.

   (i) Find the probability mass function of $X + Y$.

   (a) $k\dfrac{9}{16 \cdot 4^k}$

   (b) $(k + 1)\dfrac{9}{16 \cdot 4^k}$

   (c) $(k + 1)\dfrac{9}{16 \cdot 4^{k+1}}$

   (d) $k\dfrac{9}{16 \cdot 4^{k+1}}$

Solution:
X, Y is in the range $\{0, 1, 2, \ldots\}$.
The range of $X+Y$ is $\{0, 1, 2, \ldots\}$.

$X+Y$	$f_{X+Y}(k)$
0	$\frac{9}{16}$
1	$2\frac{9}{16\times 4}$
2	$3\frac{9}{16\times 4^2}$
$\vdots$	$\vdots$
k	$(k+1)\frac{9}{16\cdot 4^k}$

Therefore, $P(X+Y=k) = (k+1)\dfrac{9}{16\cdot 4^k}$

(ii) Find the probability mass function of $Z = \max\{X, Y\}$.

(a) $f_Z(k) = \dfrac{3(4^k-1)}{2\cdot 4^{2k}}$ for $k = 1, 2, \ldots$

(b) $f_Z(k) = \dfrac{3(4^k-1)}{2\cdot 4^{2k}}$ for $k = 0, 1, \ldots$

(c) $f_Z(k) = \dfrac{9}{16\cdot 4^{2k}} + \dfrac{3(4^k-1)}{2\cdot 4^{2k}}$ for $k = 0, 1, \ldots$

(d) $f_Z(k) = \dfrac{9}{16\cdot 4^{2k}} + \dfrac{3(4^k-1)}{2\cdot 4^{2k}}$ for $k = 1, 2, \ldots$

Solution:
The joint distribution of X and Y is given by

$$f_{XY}(x, y) = \frac{9}{16\times 4^{x+y}}$$

Let $Z = \max\{X, Y\}$.
Clearly, range of Z will be $T_Z = \{0, 1, 2, \ldots\}$.

Now, choose $k \in T_Z$. Then

Z	$f_Z(k)$
0	$\frac{9}{16}$
1	$2\frac{9}{16\cdot 4}+\frac{9}{16\cdot 4^2}$
2	$2\frac{9}{16\cdot 4^2}+2\frac{9}{16\cdot 4^3}+\frac{9}{16\cdot 4^4}$
$\vdots$	$\vdots$
k	$2\frac{9}{16\cdot 4^k}\left(1+\frac{1}{4}+\frac{1}{4^2}+\ldots+\frac{1}{4^{(k-1)}}\right)+\frac{9}{16\cdot 4^{2k}}$

Now,

$$\begin{aligned}
f_Z(k) =& 2\frac{9}{16\cdot 4^k}\left(1+\frac{1}{4}+\frac{1}{4^2}+\ldots+\frac{1}{4^{(k-1)}}\right)+\frac{9}{16\cdot 4^{2k}} \\
=& 2\frac{9}{16\cdot 4^k}\left(\frac{1-\left(\frac{1}{4}\right)^k}{1-\frac{1}{4}}\right)+\frac{9}{16\cdot 4^{2k}} \\
=& \frac{9}{16\cdot 4^{2k}}+\frac{3(4^k-1)}{2\cdot 4^{2k}}
\end{aligned}$$

Therefore, $f_Z(k)=\dfrac{9}{16\cdot 4^{2k}}+\dfrac{3(4^k-1)}{2\cdot 4^{2k}}$ for $k=0,1,\ldots$

10. Let the random variables X and Y, which represent the number of calls received by call centers A and B, respectively, in a one-hour interval follow the Poisson distribution. The average number of calls received in call centers A and B is 2 per hour and 3 per hour, respectively. Assume that X and Y are independent. If Z denotes the total number of calls received in call centers A and B, find the conditional probability $f_{Y|Z=5}(3)$. Enter the answer correct to two decimal places.
Answer: [0.33, 0.36]

Solution:

Given, X and Y denote the number of calls received by call center A and B in a one-hour interval.
Also, $X\sim$ Poisson(2) and $Y\sim$ Poisson(3) are independent.

$Z=X+Y$ represents the total calls received in a one-hour interval.

Now, $Y\mid Z=5\sim$ Binomial$\left(5,\dfrac{3}{2+3}\right)$

Therefore, $f_{Y|Z=5}(3)={}^5C_3\left(\dfrac{3}{5}\right)^3\left(\dfrac{2}{5}\right)^2=0.35$